



Bioenergetics

High-Energy Compounds

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Clinical Biochemistry

6-10-2026

Al-Ameed College of Medicine
[2026-2027]

Learning Objectives:

Upon completion of this lecture, you should be able to:

- 1- **Define** "High-Energy Compound" correctly.
- 2- **State** the ΔG° and explain the exergonic nature of ATP hydrolysis.
- 3- **Describe** the principle of energy coupling using ATP.
- 4- **Rank** common compounds by phosphoryl group-transfer potential.
- 5- **Identify** the role of Creatine phosphate as an energy buffer.
- 6- **Recognize** thioesters as a major class of energy-rich compounds.

What is a "High-Energy Compound"?

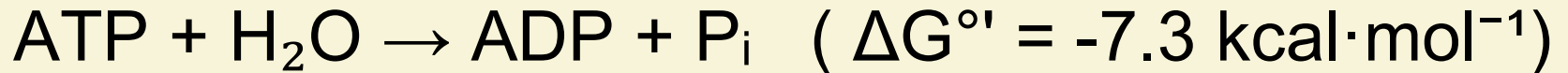
- **Not a "High-Energy Bond":** The term is a misnomer; it refers to the high negative free energy change (ΔG°) of **hydrolysis**.
- **Phosphoryl Group-Transfer Potential:** The tendency of a molecule to transfer its phosphoryl group ($\sim P$) to an acceptor, like ADP.
- The transfer potential is measured relative to the **ΔG° of ATP hydrolysis** ($-7.3 \text{ kcal}\cdot\text{mol}^{-1}$).
- A compound with a *more negative* ΔG° of hydrolysis has a *higher* phosphoryl group-transfer potential than ATP.

ATP: The Central Energy Carrier

- **Structure and Hydrolysis**

Structure: Adenine + Ribose + Triphosphate

Hydrolysis Reaction:



- **Why is $\Delta G^{\circ'}$ so Negative?**

The closely spaced negative charges on the phosphate groups at physiological pH.

- **The products (ADP and P_i) are stabilized by resonance more than the reactants.**

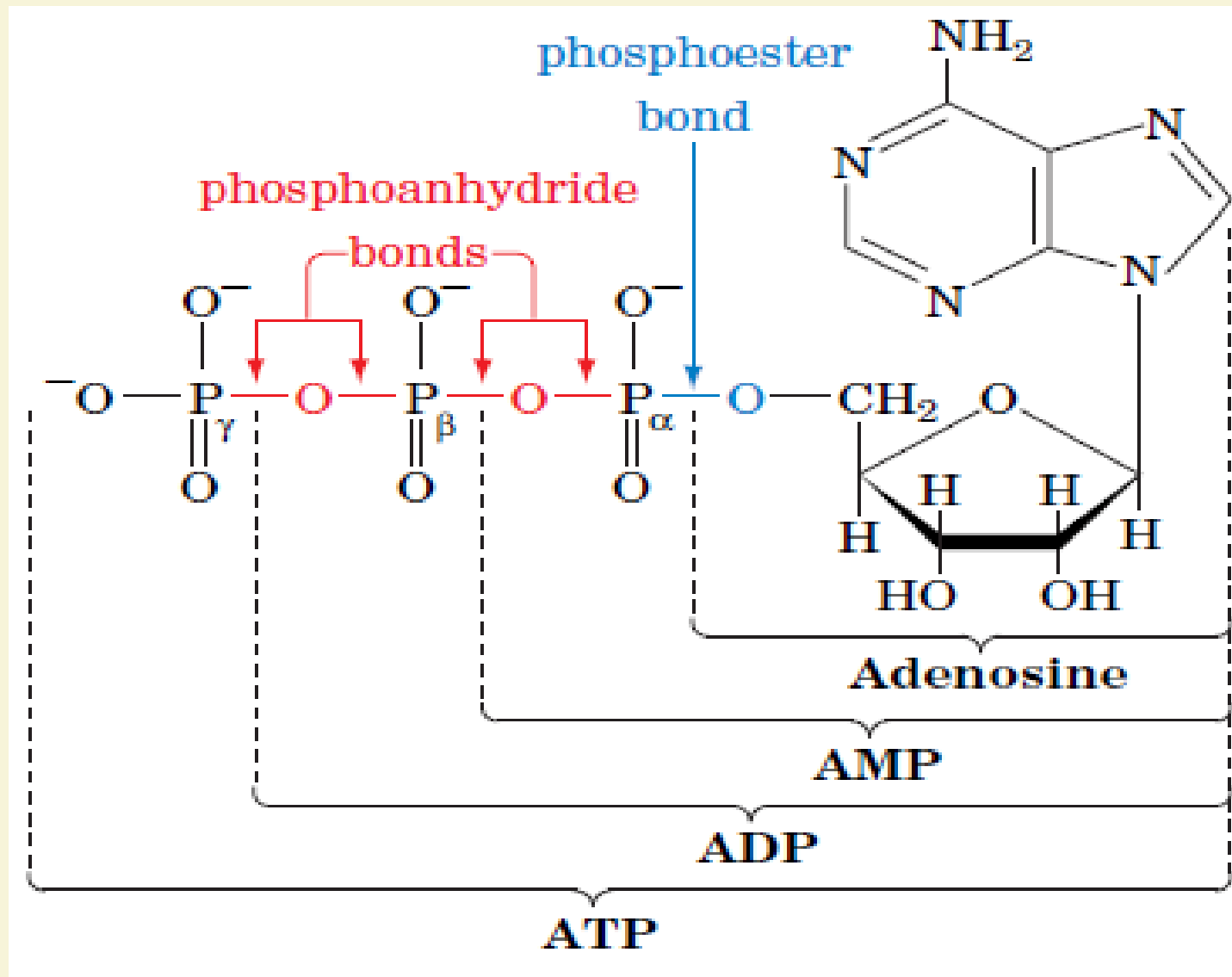


Fig.1: The structure of ATP indicating its relationship to ADP, AMP, and adenosine.

The Role of ATP in Coupled Reactions

Driving Endergonic Processes

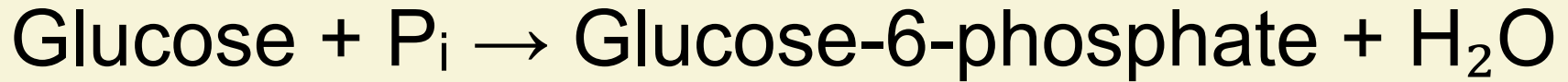
- **Principle:** ATP hydrolysis (exergonic) can be coupled to an endergonic reaction to make the overall process spontaneous.
- **Mechanism:** The phosphoryl group ($\sim P$) from ATP is transferred to a reactant molecule, forming a high-energy intermediate that is more reactive.

Overall Free Energy Change:

- $\Delta G^{\circ} \text{overall} = \Delta G^{\circ} \text{endergonic} + \Delta G^{\circ} \text{ATP hydrolysis}$
- If the sum is **negative**, the coupled reaction proceeds.

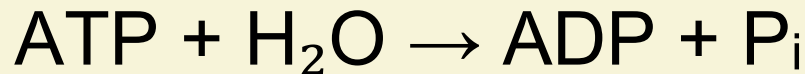
Coupled Reaction Example

- Endergonic Reaction:



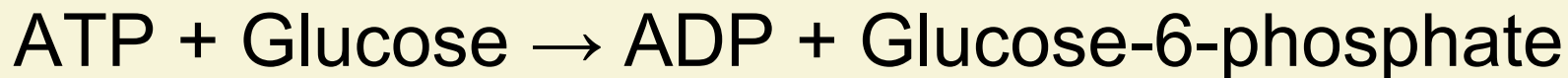
$$\Delta G^{\circ'} = +3.3 \text{ kcal}\cdot\text{mol}^{-1} \text{ (Non-spontaneous).}$$

- **Coupling with ATP:**



$$\Delta G^{\circ'} = -7.3 \text{ kcal}\cdot\text{mol}^{-1}$$

- **Overall Coupled Reaction:**



$$\Delta G^{\circ'} \text{ overall} = -4.0 \text{ kcal}\cdot\text{mol}^{-1} \text{ (Spontaneous)}$$

Phosphoryl Group-Transfer Potential Scale

- **A Thermodynamic Hierarchy**

Compounds can be ranked by their tendency to donate a phosphoryl group.

- **High-Transfer Potential Donors:**
Phosphoenolpyruvate (PEP), 1,3-Bisphosphoglycerate (1,3-BPG), and Creatine Phosphate can phosphorylate ADP to make ATP.
- **ATP occupies a middle position**, allowing it to accept $\sim P$ from high-potential compounds and donate $\sim P$ to low-potential compounds.

Q/ During the sprint, creatine phosphate regenerates ATP: Creatine phosphate + ADP $\rightarrow$ Creatine + ATP. Using creatine phosphate hydrolysis ($\Delta G^{\circ} = -10.3$) and the reverse of ATP hydrolysis (ADP + $P_i \rightarrow$ ATP + H_2O , $\Delta G^{\circ} = +7.3 \text{ kcal}\cdot\text{mol}^{-1}$), what is the overall ΔG° and what does it tell you?

A) $-3.0 \text{ kcal}\cdot\text{mol}^{-1}$; creatine phosphate has a higher transfer potential than ATP, so it can phosphorylate ADP.

B) $-17.6 \text{ kcal}\cdot\text{mol}^{-1}$; the reaction releases more energy than the sum of its parts.

C) $+3.0 \text{ kcal}\cdot\text{mol}^{-1}$; creatine phosphate cannot regenerate ATP.

D) $+17.6 \text{ kcal}\cdot\text{mol}^{-1}$; ATP must be added to drive it.

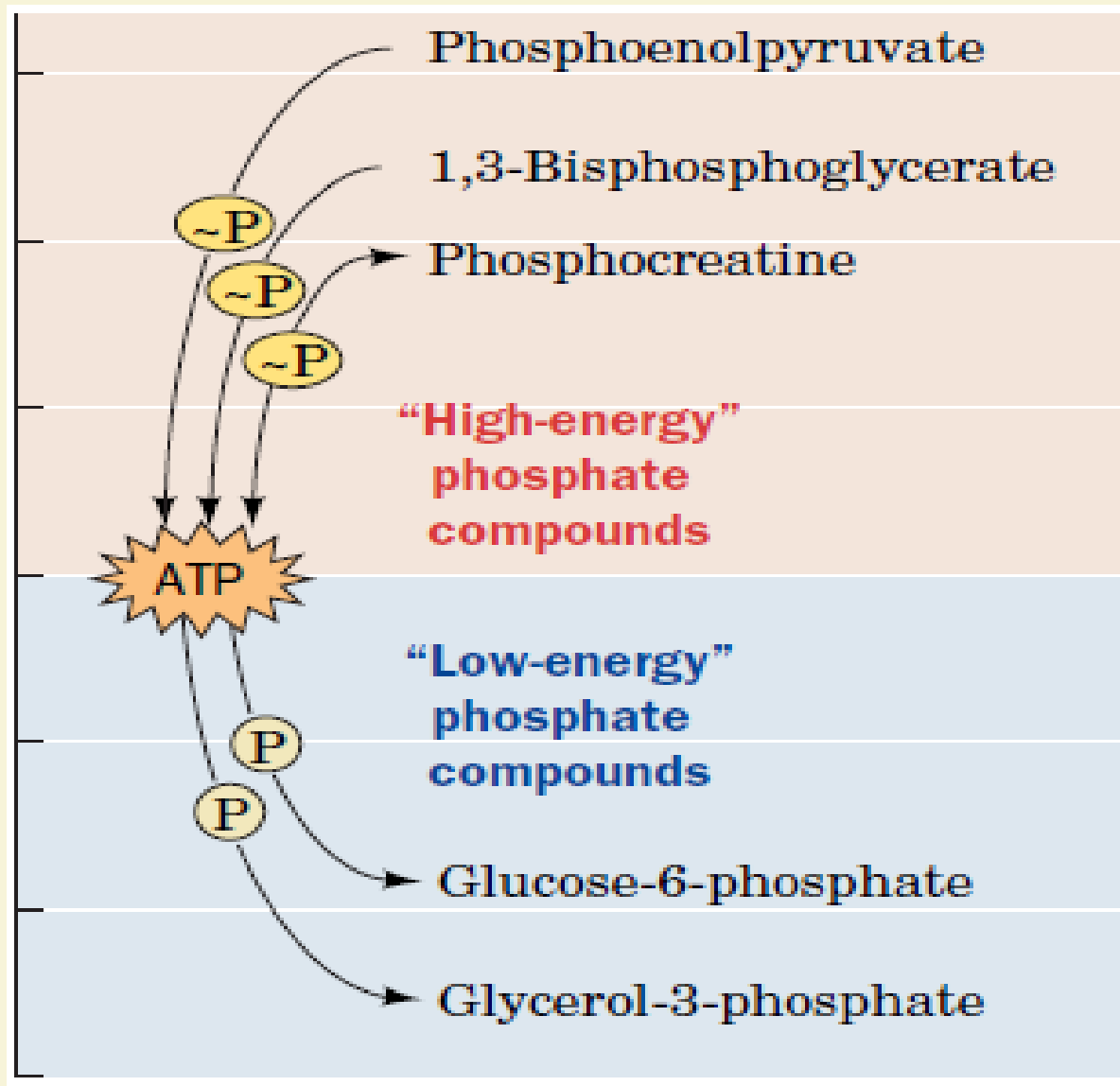


Fig.2: Position of ATP relative to “high-energy” and “low-energy” phosphate compounds. Phosphoryl groups flow from the “high-energy” donors, via the ATP–ADP system, to “low-energy” acceptors.

High-Energy Phosphorylated Compounds

- Phosphoenolpyruvate (PEP)
- Structure: An enol phosphate.
- Hydrolysis:
- $\text{PEP} + \text{H}_2\text{O} \rightarrow \text{Pyruvate} + \text{P}_i$
- $\Delta G^\circ = -14.8 \text{ kcal}\cdot\text{mol}^{-1}$
- The enol form of pyruvate is unstable and spontaneously converts to the highly stable keto form.

- **1,3-Bisphosphoglycerate (1,3-BPG)**
- Structure: An acyl phosphate.
- Hydrolysis:
- $1,3\text{-BPG} + \text{H}_2\text{O} \rightarrow 3\text{-Phosphoglycerate} + \text{P}_i$
- $\Delta G^\circ = -11.8 \text{ kcal}\cdot\text{mol}^{-1}$
- High-Energy Rationale: The acyl phosphate group has a highly polarized carbonyl carbon that is very susceptible to nucleophilic attack, and the products are stabilized by resonance.

- **Creatine Phosphate**
- A Molecular Energy Reservoir
- Function: Serves as a rapidly mobilizable reserve of high-energy phosphates in muscles and nerve tissues.
- Hydrolysis:
- Creatine Phosphate + H₂O → Creatine + P_i
- ** $\Delta G^{\circ} = -10.3 \text{ kcal} \cdot \text{mol}^{-1}$
- Recharging: When ATP is plentiful, its ~P can be used to re-synthesize Creatine phosphate via the enzyme Creatine kinase.

- **Acetyl-Coenzyme A (Acetyl-CoA)**
- Structure: An acyl group (e.g., acetyl) is linked to the thiol ($\sim\text{SH}$) group of Coenzyme A via a thioester bond.
- Hydrolysis:
- $\text{Acetyl-CoA} + \text{H}_2\text{O} \rightarrow \text{Acetate} + \text{CoA-SH}$
- $\Delta G^\circ = -7.5 \text{ kcal}\cdot\text{mol}^{-1}$ (Similar to ATP)
- High-Energy Rationale: The carbon-sulfur bond is weak.
- The products (acetate and thiol) are more stable due to greater resonance stabilization compared to the thioester reactant.

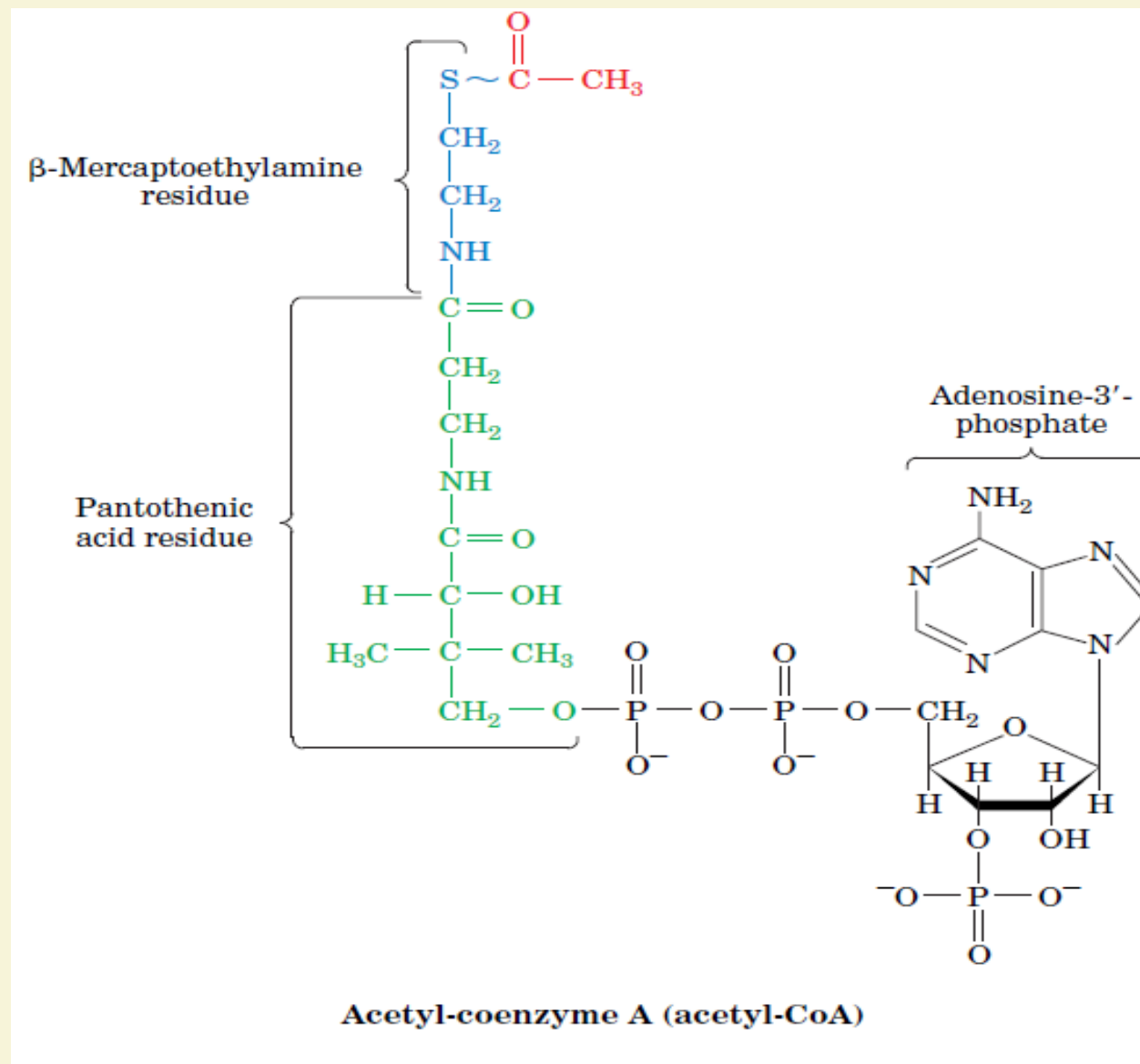


Fig.3: The chemical structure of acetyl-CoA. The thioester bond is drawn with a ~ to indicate that it is a “high-energy” bond (has a high negative free energy of hydrolysis). In CoA, the acetyl group is replaced by hydrogen.

Reference:

- Voet, D., Voet, J.G., & Pratt, C.W. (2016). *Fundamentals of Biochemistry: Life at the Molecular Level* (5th Ed.). Wiley.